Knotty

Support investigations that follow the evidence



A learning experiment in MCP became a system for evidence, review and durable decisions.

Formerly Support Ticket Triage MCP

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A learning exercise became a systems project

JUNE

Connect an assistant

MCP tools and resources
Reusable Skill instructions
Synthetic ticket tools

JULY

Model the investigation

Evidence and replies
GPT advisory calls
Diagnosis review

AUGUST

Make decisions durable

Operational SQLite
Command replay
Learning outbox

SEPTEMBER–OCTOBER

Test richer reasoning

Retrieval and ranking
Applicability assessment
Blinded comparisons

MCP (Model Context Protocol) provided the first tool interface.

The evidence changes the investigation

First report

“It does not work.”

Ask which feature failed.

More context

The campaign editor opens blank.

Classify the affected area.

New evidence

Another browser and admin fail.
The console reports ChunkLoadError.

Review the current diagnosis.

TKT-1010 / Synthetic example, condensed from the implemented lifecycle.

A workflow that can reconsider its conclusion

01

Gather evidence

Read replies.
Ask for missing facts.

02

Review diagnosis

Propose a cause.
Wait for review.

03

Apply a scoped fix

Permit the current scoped command.

04

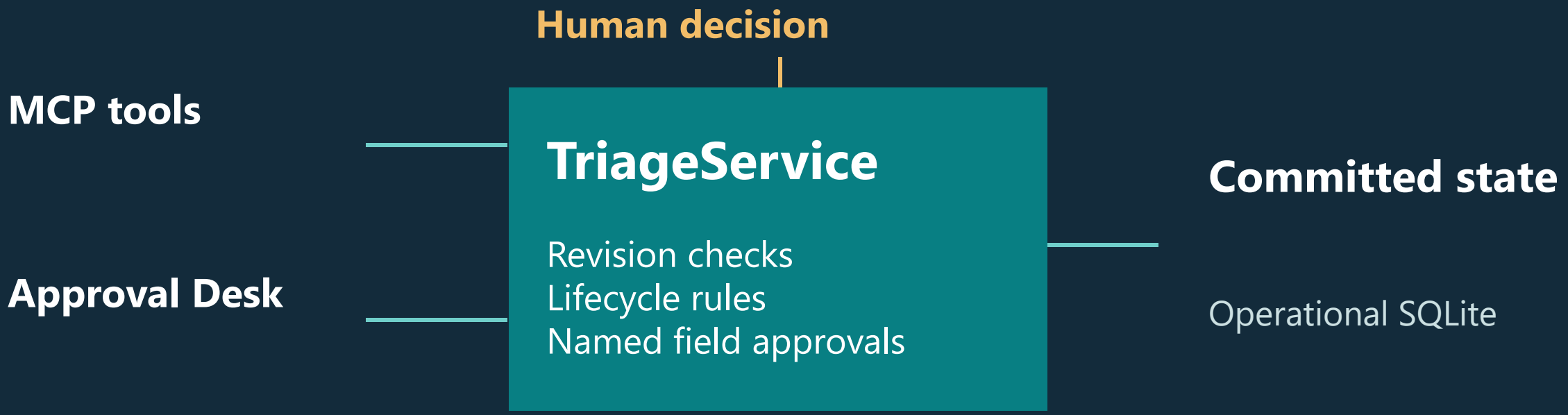
Verify the outcome

Record confirmation.
Then allow closure.



New evidence or an ineffective fix can reopen an earlier conclusion.

The model proposes. The service checks.



Ticket text is untrusted evidence.
Model output stays advisory until the command passes policy.

A lost reply should not become two decisions

The client sends an approval

One SQLite transaction

Ticket revision
Causal event
Command receipt
Learning-outbox intent

The commit succeeds.
The connection drops.

After restart, the same command ID recovers the recorded result.

Durable command identity connects transactions, history and recovery.

Investigations become reviewable knowledge

01

Completed case

Preserve the diagnosis and its evidence.

02

Candidate knowledge

Discover a pattern. Draft reusable guidance.

03

Reviewed version

Review evidence policy. Promote an immutable version.

04

Later investigation

Retrieve knowledge. Check applicability.

The knowledge base evolves through review. Model weights stay unchanged.

Hybrid search: a research path

Research components only. Fresh Luna tests used BM25, with semantic vectors inactive.

EARLIER ARTICLE LOOKUP

Substring matching

Look for the whole query in an article's title, tags or body.

Rule-linked references remain explicit.

HYBRID RESEARCH / SHADOW PATH

Knowledge objects: articles, known causes, playbooks

Searchable sections and diagnostic fields preserve source identity.

Lexical / BM25

Rank matching terms.
Useful for exact error names.

Semantic / embeddings

Optional research channel.
Production benefit unproven.

Candidate union + explicit references

B4 ranks context. B5 assesses whether a match applies.

A retrieved match can still explain the wrong problem

RAG: retrieval-augmented generation

Shadow research only. Fresh Luna tests used lexical retrieval.

RETRIEVE

Articles, known causes and diagnostic playbooks

Lexical; semantic optional

SELECT

Rank candidates and bound the context

Existing B4 selection

ASSESS

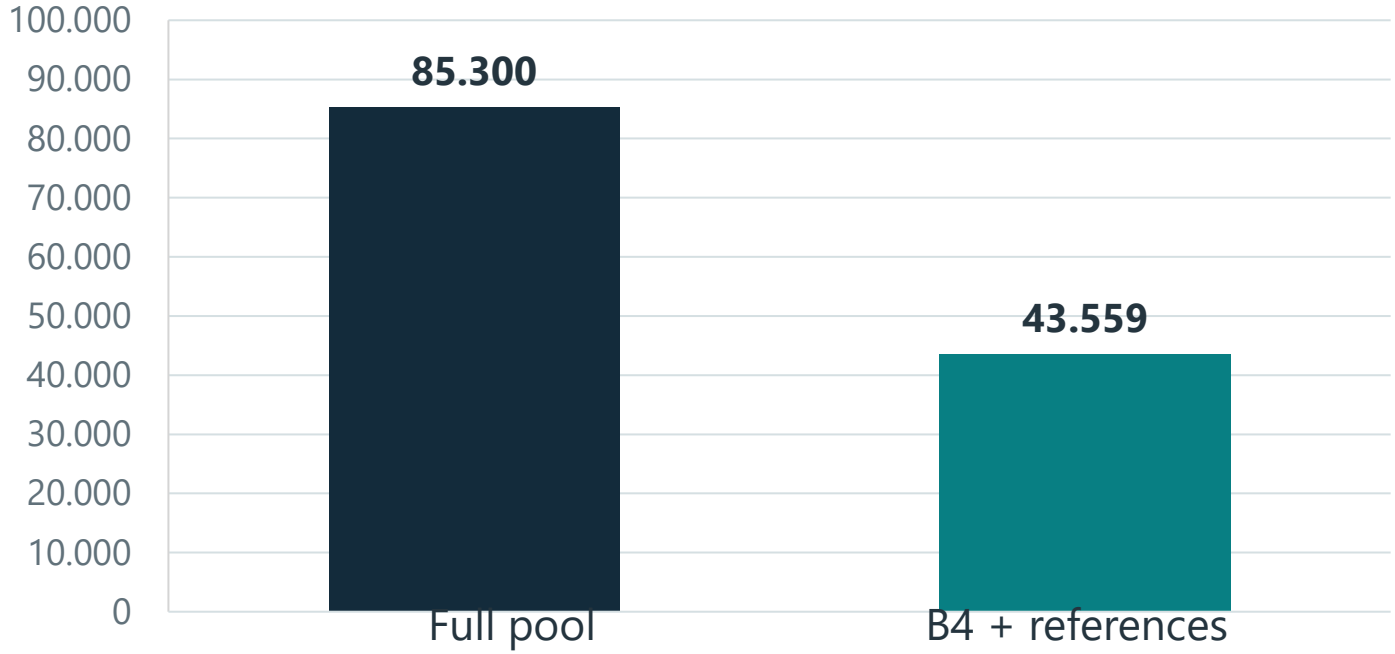
Check applicability, then synthesize the next step

B5 reasoning and H5b mapping

Example: signature verification fails. A delivery-latency cause is irrelevant.

Four synthetic cases, fewer input tokens

API input tokens: four paired synthetic cases / eight outputs



48.9% fewer input tokens

8/8 labeled relevant pairs retained (resource/case)

Human quality review pending
No proven quality or billing gain

Main run: SIMPLE 26/26 valid, B5/H5b HYBRID 26/26 valid.
Ranking follow-up: a smaller context. Diagnostic quality and billing savings remain unproven.

The engineering skills grew with the questions

PROJECT DECISION	PRACTICE DEVELOPED
Connect an assistant to tools	MCP contracts, Skills and bounded API integration
Separate proposals from authority	Domain modeling and shared service boundaries
Recover a committed decision	SQLite transactions, replay and failure analysis
Reuse knowledge and test reasoning	RAG, provenance and reproducible evaluation

Matias: direction, requirements and review, with substantial AI collaboration.

Next milestone: diagnosis quality

01

Complete the review

Review blinded pairs for diagnosis and evidence support.
Rate uncertainty and next actions.

02

Preserve more reasoning

Map useful next actions.
Connect evidence gaps to recognized requirements.

03

Test the full investigation

Exercise stateful shadow steps, changing evidence and recovery
with matched cases.

The next evidence gate informs whether richer reasoning earns operational use.

Roadmap: a durable diagnostic workbench

A support investigation that keeps its evidence, explains its proposals and reuses reviewed knowledge.

INTEGRATION

Read-only help-desk adapters
Provider mapping and reconciliation
Real identity and reviewer attribution

HARDENING

Backup and restore drills
Readiness reporting and packaging
Stronger failure and quality evaluations

Earlier tests establish working components and bounded experimental results.
Full comparative testing is still in progress.